

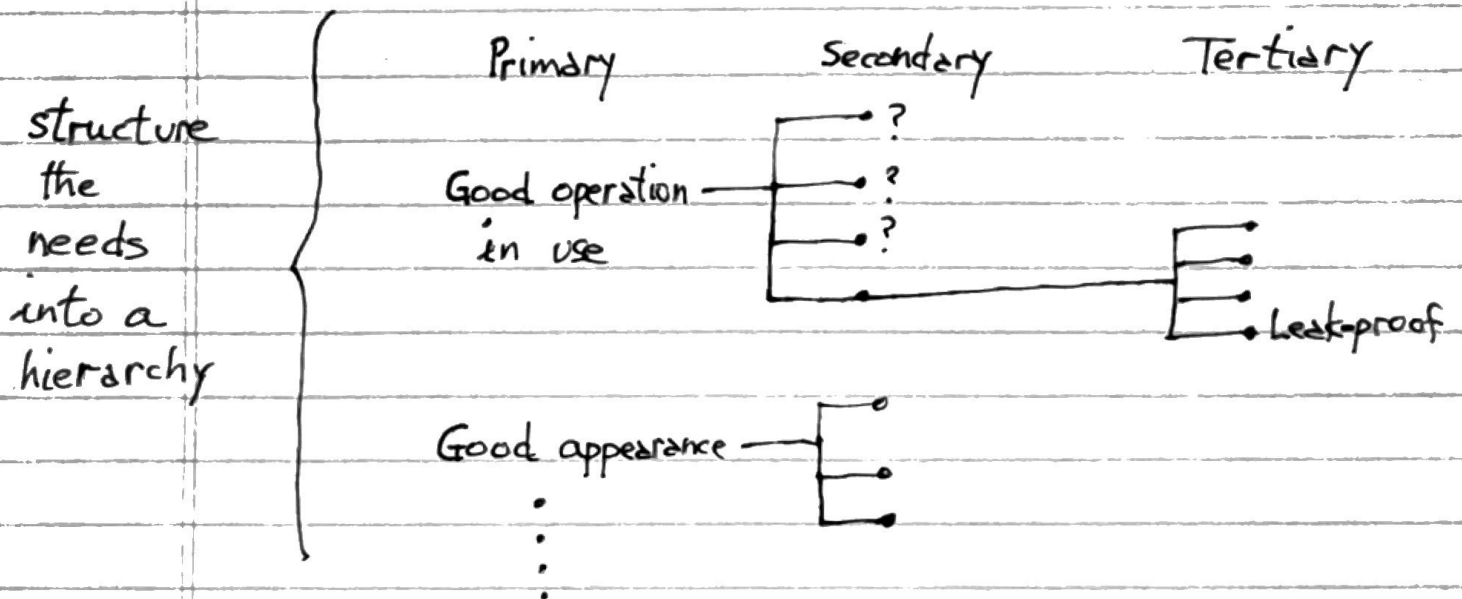
TIM 172A, MOT-I, Lecture # 9 (10/23/25)

Agenda

1. Complete the HQA process (with example).
done; see above
2. HW #5
3. Midterm (mid-term) will be handed
out in the classroom, on Thurs. 10/30/25.

L #9 starts here

(Car Door)



As a result of Step 1

① Customer Needs	② Importance	Scale: 1-10
Easy to close	7/10	
⋮	⋮	
Leak-proof	9/10	
Isolate road noise	6/10	
⋮	⋮	

Step 2: (Engineering) Make a prioritized list of technical metrics (or requirements), ③, and assess the importance of each metric, ④, using a convenient scale

Technical Metrics ③					
④	Importance	Energy to close door (lb-ft)	...	Peak closing force (lb.)	Noise reduction (dB)
		6/10		7/10	9/10

← scale of "1-10."

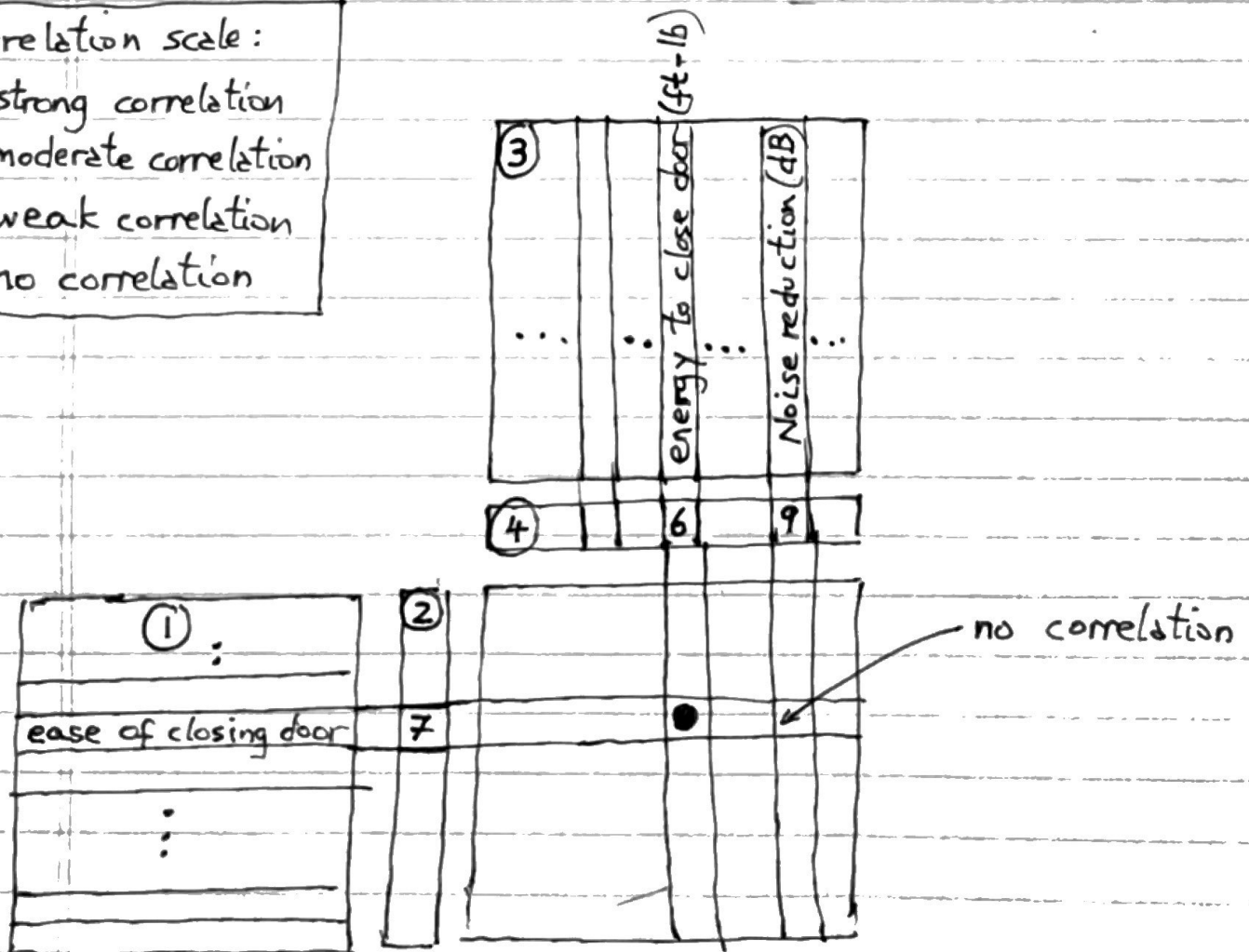
Step 3 : Correlate customer needs & technical metrics (or engineering requirements) using a convenient scale.

The result is a correlation matrix, ⑤.

for example : The customer need, "ease of closing door" is strongly correlated to the technical metric "energy to close door"

Correlation scale :

- strong correlation
- moderate correlation
- △ weak correlation
- no correlation



Step 4 : There are dependencies between the technical metrics

e.g. $\text{Energy} = (\text{force}) \times (\text{distance})$
 $\Rightarrow$ "energy required to close door"
is positively correlated to
"force to close door"

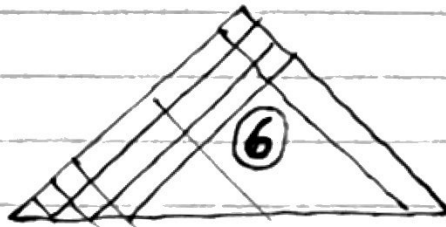
	TM1	TM2	TM3	
Technical Metric 1	+	-		
TM2	-	+		
TM3			+	(6)
⋮				

} symmetric matrix, which is symmetric about the diagonal:
therefore, we only need "half" of the matrix

4(a) : Correlate the technical metrics to each other using a convenient scale :

+ : strong positive correlation
- : strong negative correlation
: no correlation
(see above)

4(b) Place the "half" of the matrix, which is above the diagonal, (6), on top of (3).



HQ template is available at qfdi.org web-site

...	Energy to close door	Noise reduction
-----	----------------------	-----------------

(4)

Product A	Product B	...	Our Door
4	7	8	

Scale of 1-10

(1)	...
Ease of closing door	door

(2)

(5)			
-----	--	--	--

Product A	(8)	26	16
Product B		33	22
...			
Our product		30	6
Units of measurement			

ft-lb

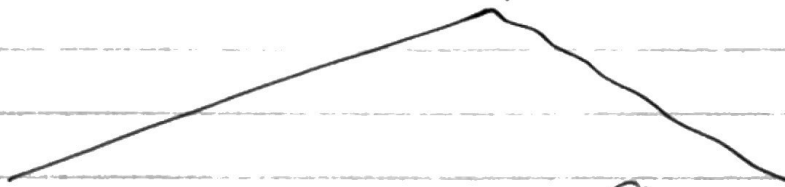
dB

Step 5 : Benchmarking

Assess a set of competing products from the viewpoint of the customer (customer benchmarking), (7), and from a technical (engineering) viewpoint (technical benchmarking), (8), using a convenient scale for customer benchmarking, and engineering units of measurement for technical benchmarking.

[see previous page; or see above]

Step 6 : Set targets for customer needs & technical metrics for the new intended product



(9) Target customer specs

- Low cost 7/10
- Appearance 4/10
- Easy to close 8/10

...

(10) Target technical specs.

Force to close door : 15 lb.

Energy to close door : 20 ft-lb

Noise reduction : 6 dB

...

How would a more complete (with more details) HOQ for a car door look?

See the "Car Door" HOQ handout on Canvas

You can present the HOQ information in one of two ways:

create an "actual" house (as in today's lecture); template is available on the "gfdi.org" web-site

still follow the process in L#9 (today), & then present the results in a manner similar to "Product Specifications" chapter of the textbook:
"bicycle suspension fork"

HW #5, Prob 1

2. HW # 5, Prob # 2

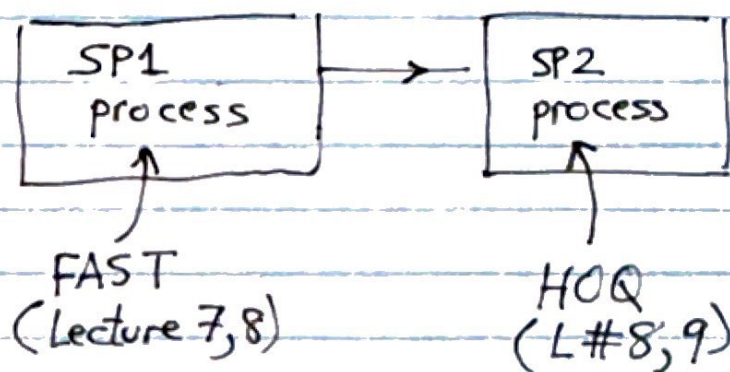
Prob. as stated : 'Create a HQQ ... -

SPSP : Step 1 : Define the (real) problem

SP1 : Dissect the product

SP2 : Construct the HQQ for
the dissected product

Step 2 : Plan or Process



Step 3 : Execution